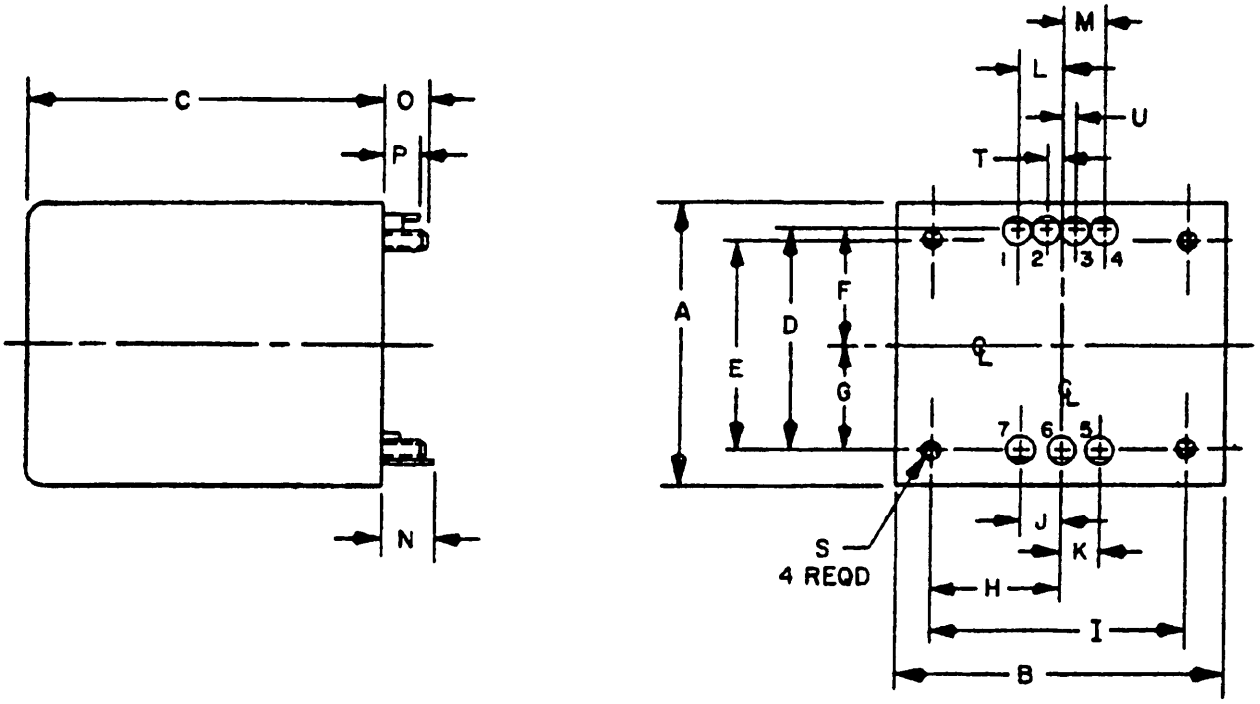
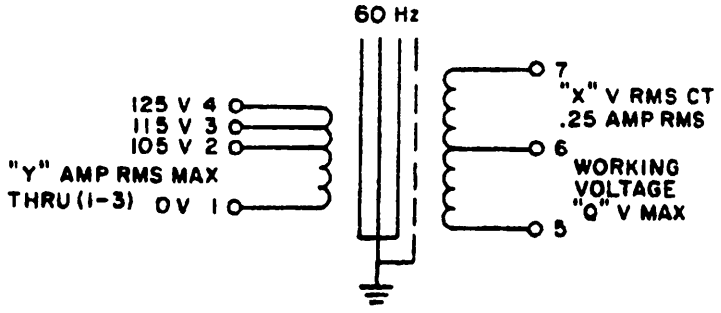
MILITARY SPECIFICATION SHEET

TRANSFORMERS, POWER, STEPUP, TF4RO3

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.





WORKING VOLTAGE 175 V MAX
MAX ALTITUDE: 10,000 FT

CIRCUIT DIAGRAM AND MARKING

Ⓐ denote changes

FIGURE 1. Dimensions and configurations.

FSC 5950

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INCHES	MM	INCHES	MM	INCHES	MM
.016	.41	.656	16.66	2.875	73.02
.030	.76	1.218	30.94	3.000	76.00
.047	1.19	1.344	34.14	3.188	80.98
.125	3.18	1.375	34.92	3.312	84.12
.188	4.78	1.500	38.10	3.375	85.72
.25	6.4	1.531	38.10	3.688	93.68
.375	9.52	1.656	38.89	3.938	100.03
.438	11.13	1.719	43.66	4.062	103.17
.500	12.70	2.031	51.59	4.312	109.52
.562	14.27	2.438	61.93	4.500	114.30
.625	15.88	2.688	68.28	5.062	128.57
		2.750	69.85	5.500	139.70

Dash number	Type designation	A +.000 -.125	B +.000 -.125	C +.000 -.188	D ±.125	E ±.047	F ±.125	G ±.125	H ±.125	I ±.047
-01	TF4R03KB	3.375	3.938	4.312	2.750	2.438	1.375	1.218	1.5	
-02	TF4R03LB	3.688	4.312	4.500	2.875	2.688	1.531	1.344	1.656	3.31
-03	TF4R03NB	4.312	5.062	5.500	3.188	3.312	1.719	1.656	2.031	4.06

Dash number	M ±.030	N (MAX) Terminals 5, 6, & 7	O (±.125) Mounting studs	P (MAX) Terminals 1 thru 4	S Mounting studs	T ±.016	U ±.016
-01	.562	.438	.500	.438	.190-32 UNF-2A	.188	.188
-02	.562	.656	.500	.438	.190-32 UNF-2A	.188	.188
-03	.562	.656	.625	.438	.25-20 UNC-2A	.188	.188

(A)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Electrical values shall be marked as specified in table I.

FIGURE 1. Dimensions and configurations - Continued.

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TABLE I. Electrical characteristics.

Dash no.	Secondary voltage CT	Secondary current	Primary current (max)	Secondary working voltage (max)	Secondary DWV	Full-wave rectifier
	Volts rms	Amperes rms	Amperes rms	Volts	Volts rms	
01	700	.25	1.05	495	2400	5U4GB
02	1100	.25	1.52	780	3200	5R4WGB
03	1600	.25	2.29	1065	4200	5R4WGB

TABLE I. Electrical characteristics - Continued.

	No load			Rated load		
Dash no.	Primary current (max)	Primary power (max)	Secondary voltage (max)	Filter inductance	Secondary voltage ±4%	Weight (max)
	Amperes rms	Watts	Volts	Henrys	Volts	Pounds
01	.35	4.7	743	5.0	700	7.5
02	.37	6.0	1163	9.5	1100	11.0
03	.65	10.0	1673	4.0	1600	16.5

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage, 60 Hz ±10 percent.

(1-2): 105 volts.

(1-3): 115 volts.

(1-4): 125 volts.

Secondary voltage: See table I.

Secondary current: See table I.

Working voltage:

(1-4): 175 volts maximum.

(5-7): See table I.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal encased.

Material: Steel.

Terminals: Solder lug, No. 18 AWG.

Height: See figure 1.

Weight: See table I.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage (at sea level):

(1-4): 1500 Vrms.

(5-7): See table I.

Electrical characteristics:

No load: With 115 volts, 60 Hz across (1-3):

Current in (1-3): See table I.

Power in (1-3): See table I.

Voltage across (5-7): See table I.

Voltage across (1-2): 105 volts ±1 percent.

Voltage across (1-4): 125 volts ±1 percent.

Rated load: With 115 volts, 60 Hz across (1-3), and an inductive filter and full-wave rectifier (see table I) across (5-7), the voltage across (5-7) shall be as specified in table I.

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Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive, with terminals 4 and 5 connected.

Temperature rise: 40°C maximum with 105 volts, 54 Hz across (1-2) at an ambient temperature of 65°C, and (5-7) loaded with the full-wave rectifier and inductive input filter as specified in table I.

Marking location: See figure 1.

Part number: M27/302-(dash number from table I).

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, AL, ME
Navy - MC, AS, CG
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5950-0616-1)

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